

IoT Based Smart Pillbox & Vital Health Monitoring

Your Smart Pal for Meds and Monitoring!

Group Members (E&TC-A)

Roll No.	Name
51.	Siddhi Mahajan
52.	Vedant Mali
53.	Shourya Mall
54.	Krishna Maurya
55.	Rudra Mayekar

**A Report On Mini Internship Project
Under Guidance of**

Pooja Sarode

Assistance Professor

ES&H, TCET.

A.Y. 2025-26

ABSTRACT

Medication non-adherence is a major challenge in healthcare, particularly among elderly patients and individuals suffering from chronic diseases. Forgetting doses, overdosing, and lack of remote monitoring can lead to serious health complications.

The **MedLink IoT** is a smart, IoT-based automated pill dispenser designed to address these issues. The system dispenses medicines at scheduled times, monitors basic health parameters (Heart Rate & SpO2), and allows caregivers to track medication adherence remotely through a mobile application and cloud dashboard. By integrating embedded systems, sensors, and IoT platforms, this project provides an affordable, reliable, and user-friendly healthcare solution.

1. INTRODUCTION

Medication management plays a crucial role in effective healthcare treatment. According to the World Health Organization, nearly 50% of patients do not take medicines as prescribed, which can result in treatment failure and increased healthcare costs. This problem is more severe for elderly people and patients with memory-related disorders such as Alzheimer's disease.

Traditional pillboxes rely on human memory and do not provide confirmation of medicine intake or remote monitoring. With advancements in embedded systems and Internet of Things (IoT) technology, healthcare devices can now be made smarter, more connected, and more reliable.

The MedLink IoT system aims to automate medicine dispensing, ensure accurate timing, and provide real-time monitoring and alerts, thereby improving medication adherence and patient safety.

2. SYSTEM OVERVIEW & ARCHITECTURE

The project operates on a three-layer architecture: **Input Layer**, **Processing Layer**, and **Output Layer**.

2.1 Hardware Components

- **Arduino Uno:** Acts as the primary microcontroller ("the brain") for interfacing with sensors and controlling the motors.
- **ESP32 Module:** Handles Wi-Fi connectivity and cloud communication, enabling the IoT features.

- **MAX30102 Pulse Sensor:** Measures real-time Heart Rate (BPM) and SpO2 levels to monitor patient health.
- **IR Obstacle Sensor:** Detects the physical dispensing of the pill to verify adherence.
- **Servo Motor (SG90):** Rotates the revolver mechanism to dispense pills at the precise scheduled time.
- **OLED Display (0.96"):** Displays real-time status, time, and alerts to the user.

2.2 Software Architecture

The system uses a **Progressive Web App (PWA)** built with **React.js** for the frontend and **Google Firebase** for the backend.

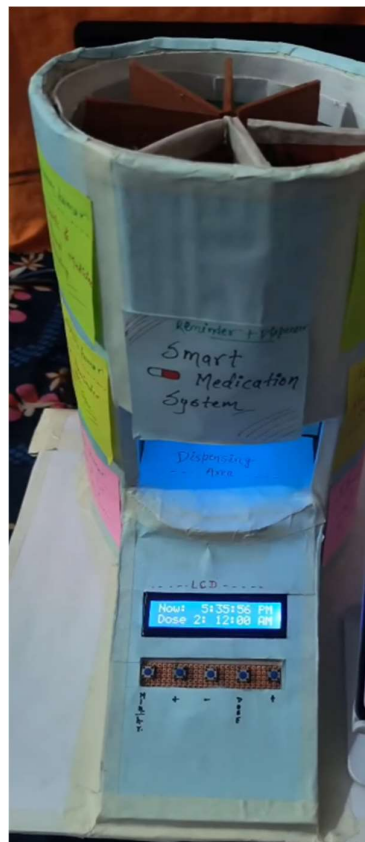
- **Frontend:** A responsive dashboard for scheduling and monitoring.
 - **Backend:** Handles authentication and real-time database syncing.
 - **Firmware:** Developed in C++ using Arduino IDE.
-

3. MECHANICAL DESIGN

A **revolver-style pill dispensing mechanism** is used, consisting of an outer casing and a rotating drum with eight compartments.

- **Mechanism:** The Servo Motor rotates the drum at fixed angles (45 degrees).
- **Dispensing:** When the compartment aligns with the bottom opening, gravity dispenses the pill.
- **Verification:** An IR sensor located at the chute confirms the pill has fallen.

Final Mechanical Prototype of the MedLink Pillbox showing the dispensing area and LCD interface.



4. RESULTS AND DISCUSSIONS

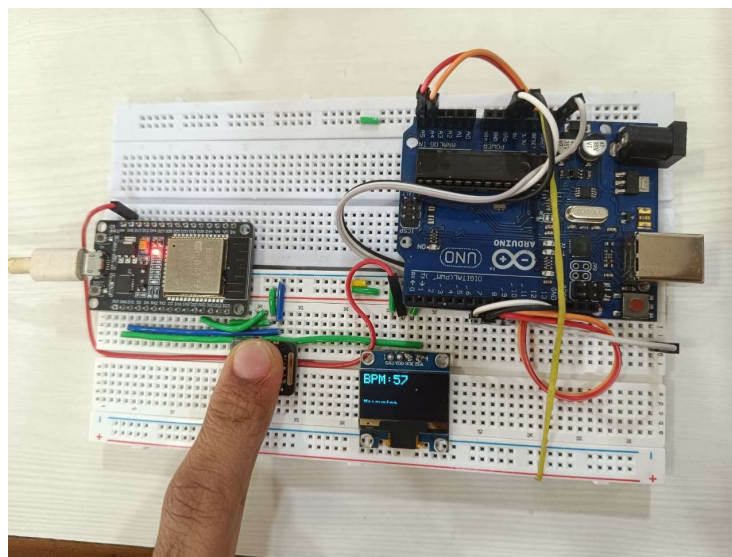
The MedLink IoT system was successfully implemented and tested under different scenarios.

4.1 Hardware Testing & Sensor Interfacing

We verified the connections and data accuracy of the sensors.

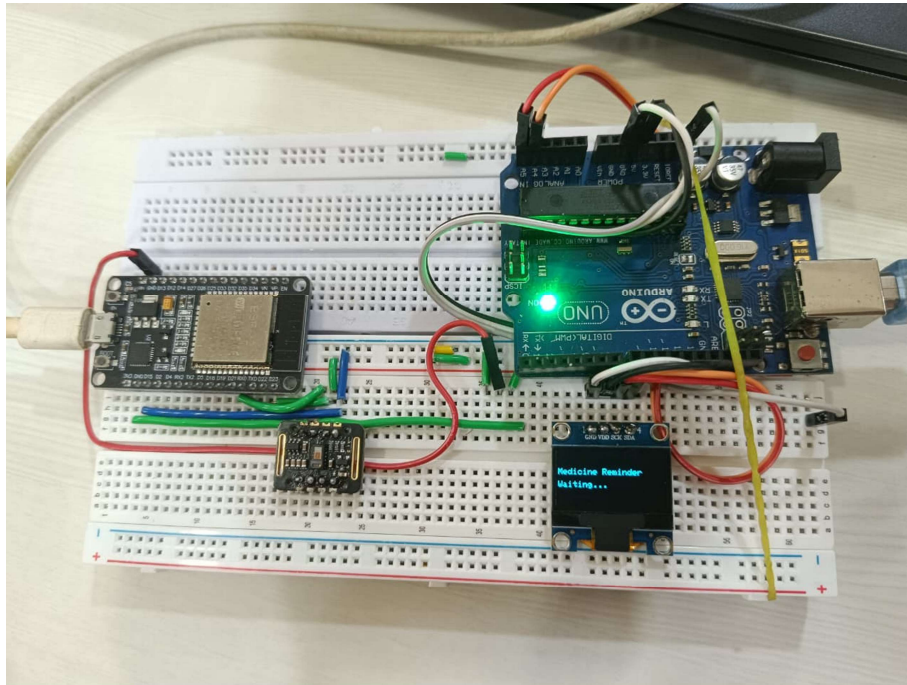
- **Heart Rate Monitoring:** The Pulse Sensor was successfully interfaced with the Arduino/ESP32. As seen in the testing phase, the sensor accurately detected a live heart rate (e.g., 57 BPM - 77 BPM) and displayed it on the OLED screen.

1. Interfacing the Pulse Sensor with ESP32 and displaying live BPM on the OLED screen.

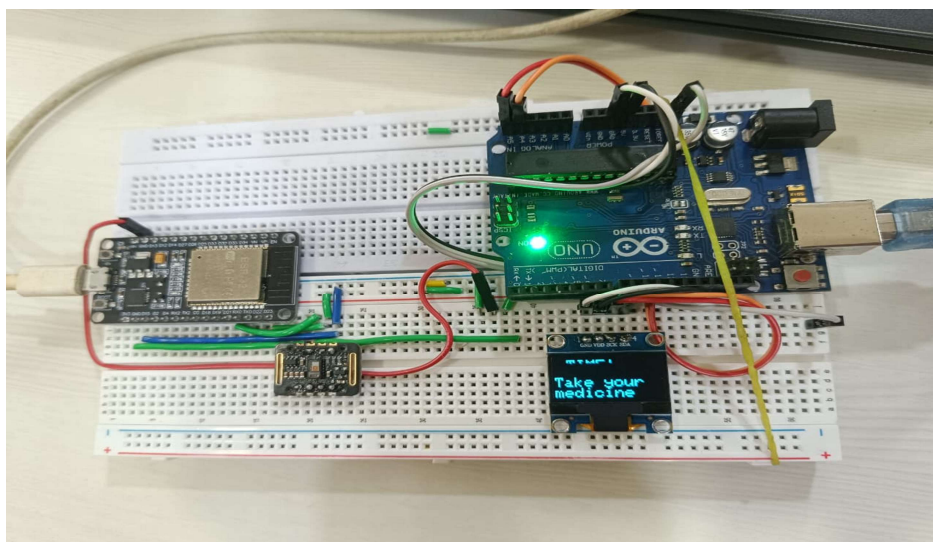


- **Alert System:** The OLED display provided clear status updates. In "Idle Mode," it shows "Medicine Reminder Waiting," and during a scheduled dose, it alerts the user to "Take your medicine."

1. OLED Display showing the system in 'Waiting' mode.



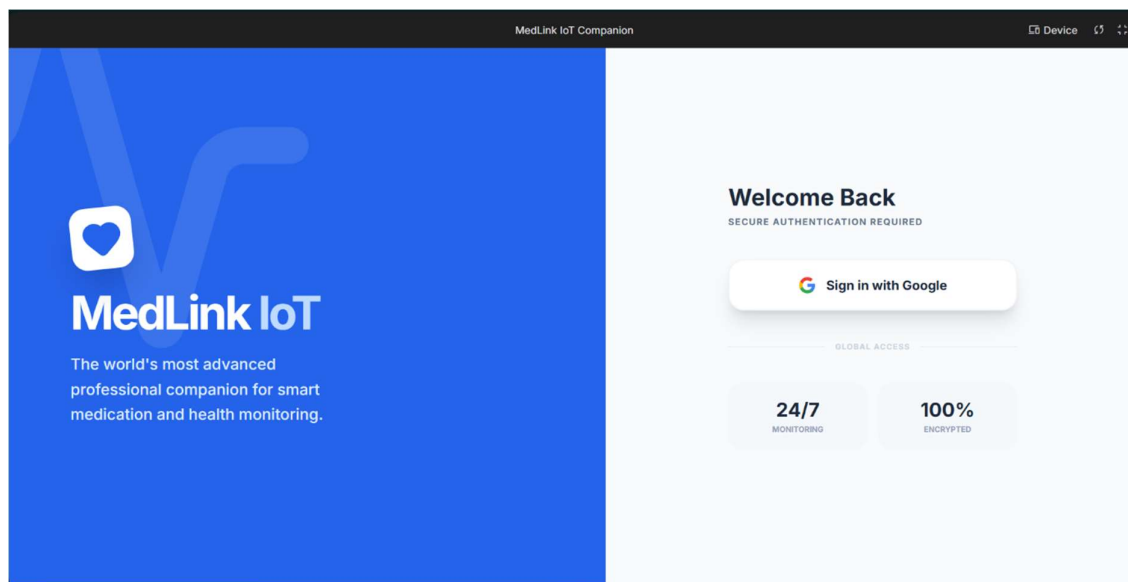
2. OLED Display triggering an active 'Take your medicine' alert.



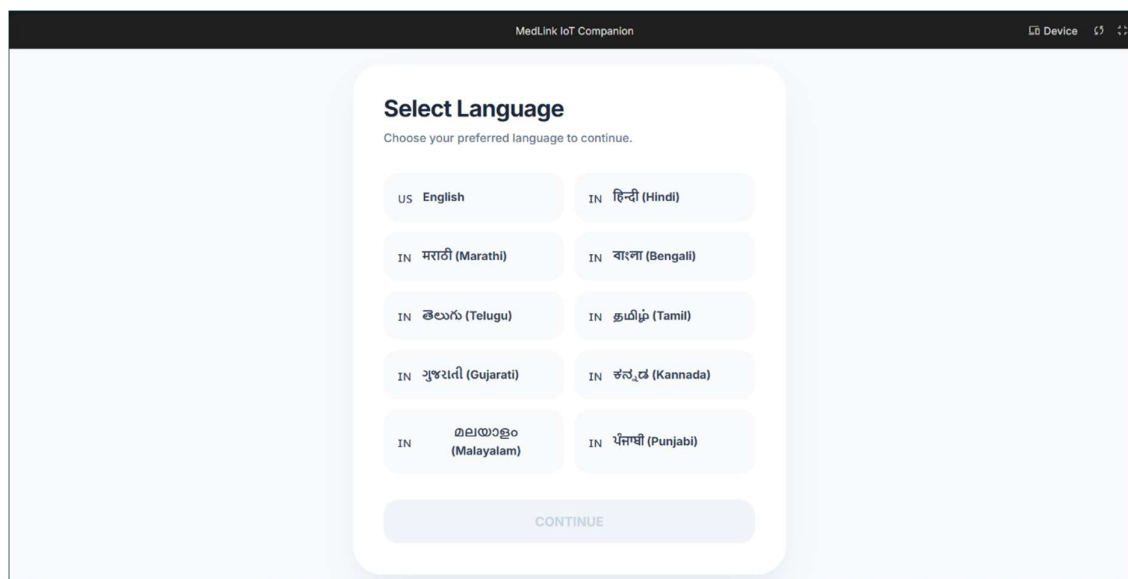
4.2 Software Interface (Screenshots)

The web application provides a user-friendly interface for monitoring and control.

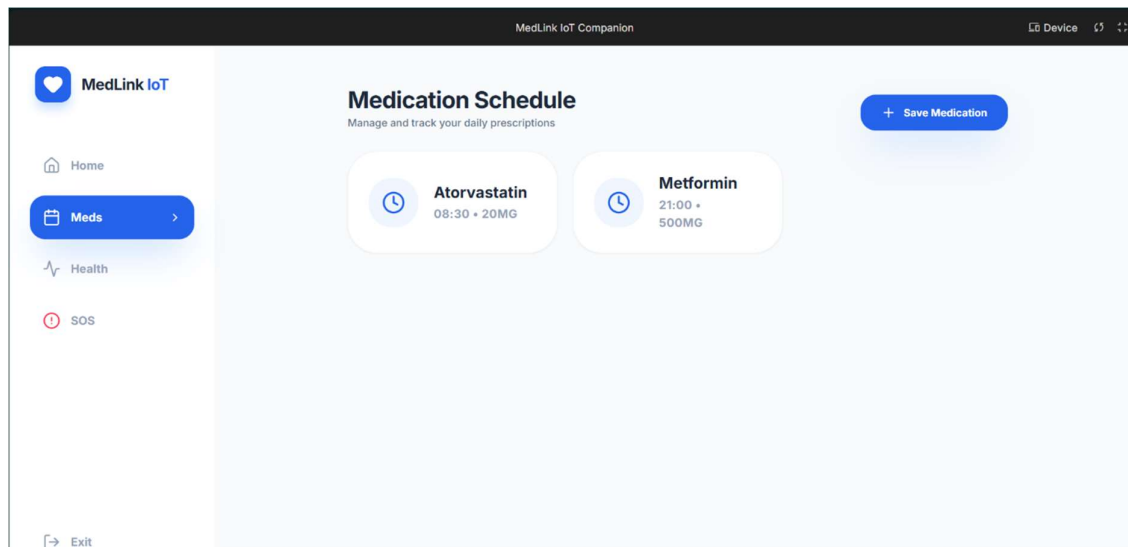
1. Secure Login Screen using Google Authentication.



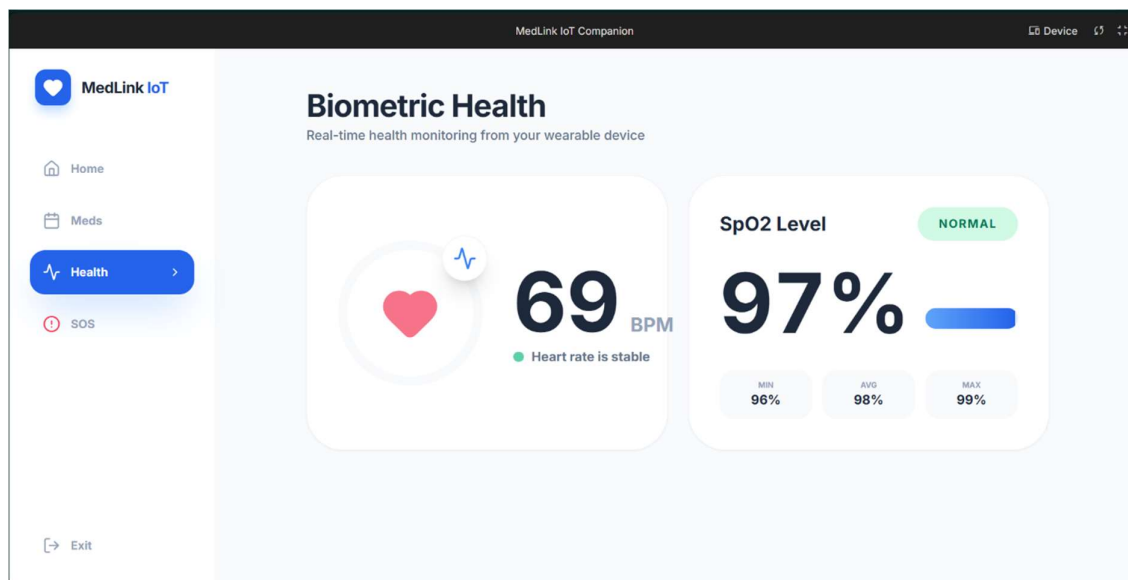
2. Language Selection Screen supporting Indian regional languages.



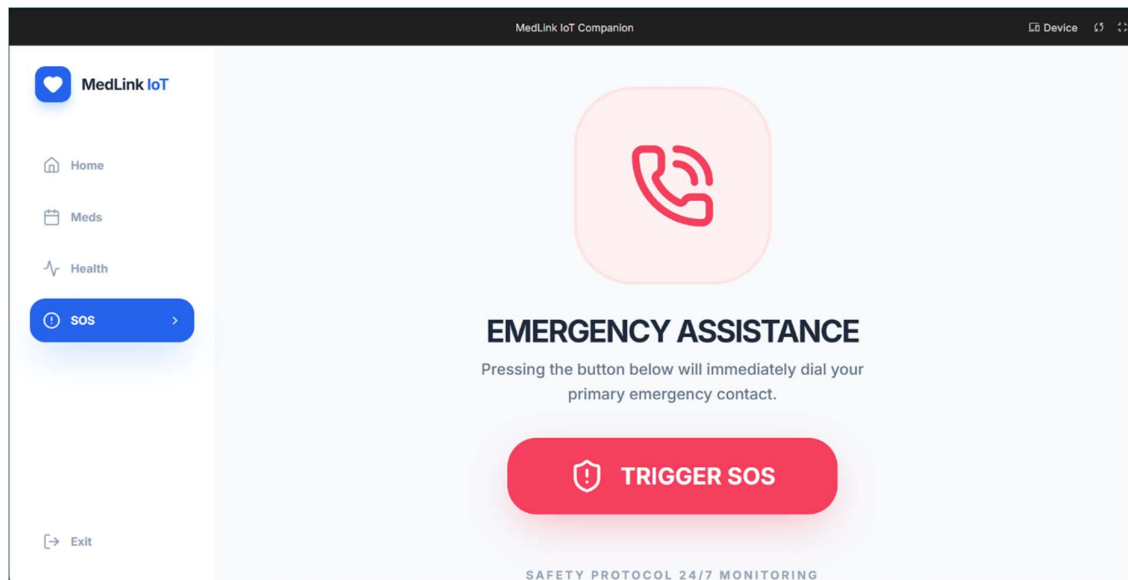
3. Main Dashboard showing "Next Dose" countdown and device status.



4. Biometric Health Tab visualizing real-time Heart Rate & SpO2.



5. Emergency SOS Screen for immediate assistance.



5. CONCLUSION

The **MedLink IoT** system successfully demonstrates how IoT and embedded systems can be used to improve medication adherence and healthcare monitoring. The project provides an affordable, automated, and user-friendly solution that enhances patient independence while allowing caregivers to monitor remotely.

6. FUTURE SCOPE

1. **Camera Integration:** Adding a mini-camera to visually confirm pill intake.
2. **Biometric Security:** Implementing fingerprint authentication to prevent unauthorized access.
3. **Multi-User Support:** Expanding the system to handle schedules for multiple patients.
4. **EHR Integration:** Linking data with Electronic Health Records for doctor access.

7. REFERENCES

1. World Health Organization - Medication Adherence Reports.
2. ESP32 Technical Reference Manual.
3. Blynk IoT Platform Documentation.
4. Arduino Official Documentation.